FFFA Donor Management MVP — System Design

Version: v1.2 (20 Oct 2026, includes FFFA's feedback) · **Status:** Reviewed by Systems lead · **Source of truth:** PRD v2 + FFFA feedback of 20 Oct (to be folded into PRD v3) · **Owner:** Systems workstream

This document describes how we build Phase 1 (DM1–DM10) by 28 Oct 2026. Anything not in PRD v2 or FFFA's 20 Oct feedback is out of scope unless the team agrees to add it.

What changed in v1.2 (FFFA feedback, 20 Oct): campaign on donations; organisation + contact person; major donor rule ($5,000+/yr or 3+ years, or tagged by staff); one-off donors lapse after 6 months; impact unit S$10 = 1 meal for one elderly person; multiple grant/sponsor due dates per donor; in-kind donations; failed monthly payments; "mark as sent" keeps the sent message on the donor's profile; import matching across Giving.sg and GIVE.asia; a documented path to add NRIC/UEN later. Schema changes land as migration `002` (001 is merged).

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1. Goals and non-goals

Goals (Phase 1)

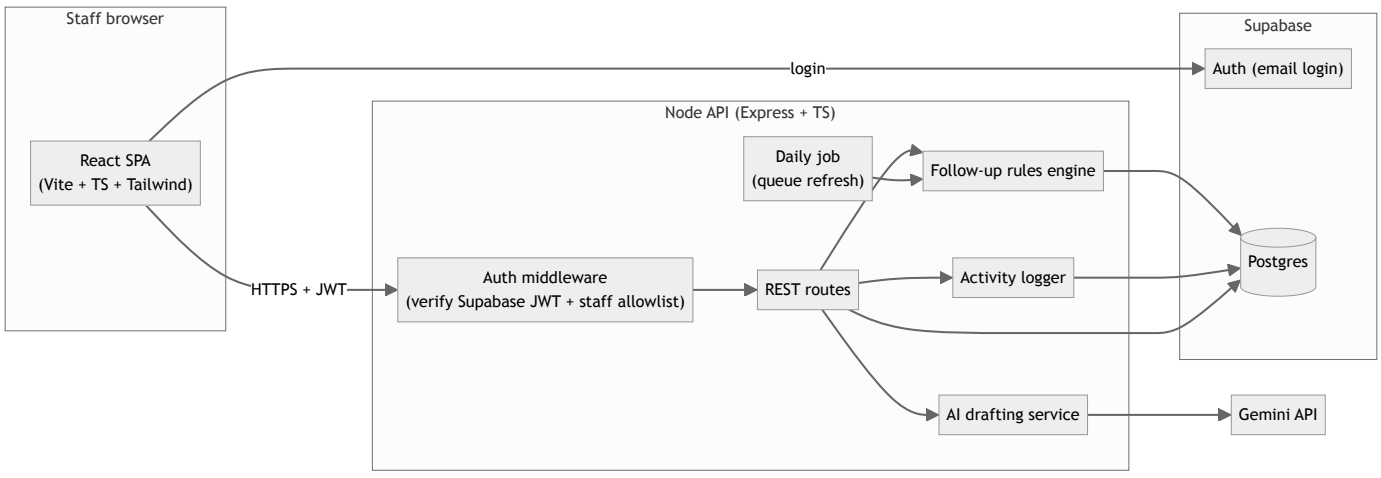
- One place where FFFA staff keep every donor, donation and relationship note (DM1, DM3, DM8).
- Follow-ups surface automatically from agreed rules instead of staff memory (DM9, DM4).
- Staff can generate an AI-drafted, personal follow-up, edit it and approve it; nothing sends automatically (DM10).
- Search, filters, dashboard, CSV import, staff-only access and an activity log (DM2, DM5, DM6, DM7).

Non-goals (Phase 1) — from PRD v2 "Out of scope"

- Sending email/WhatsApp from the system, newsletters, campaign ads, donor portal, payments.
- Volunteer or beneficiary management (only an "also volunteers" tag on donors).
- IRAS submission (but donation records stay structured so a later export is easy).
- NRIC/FIN/UEN or any sensitive attributes (age, income, ethnicity).

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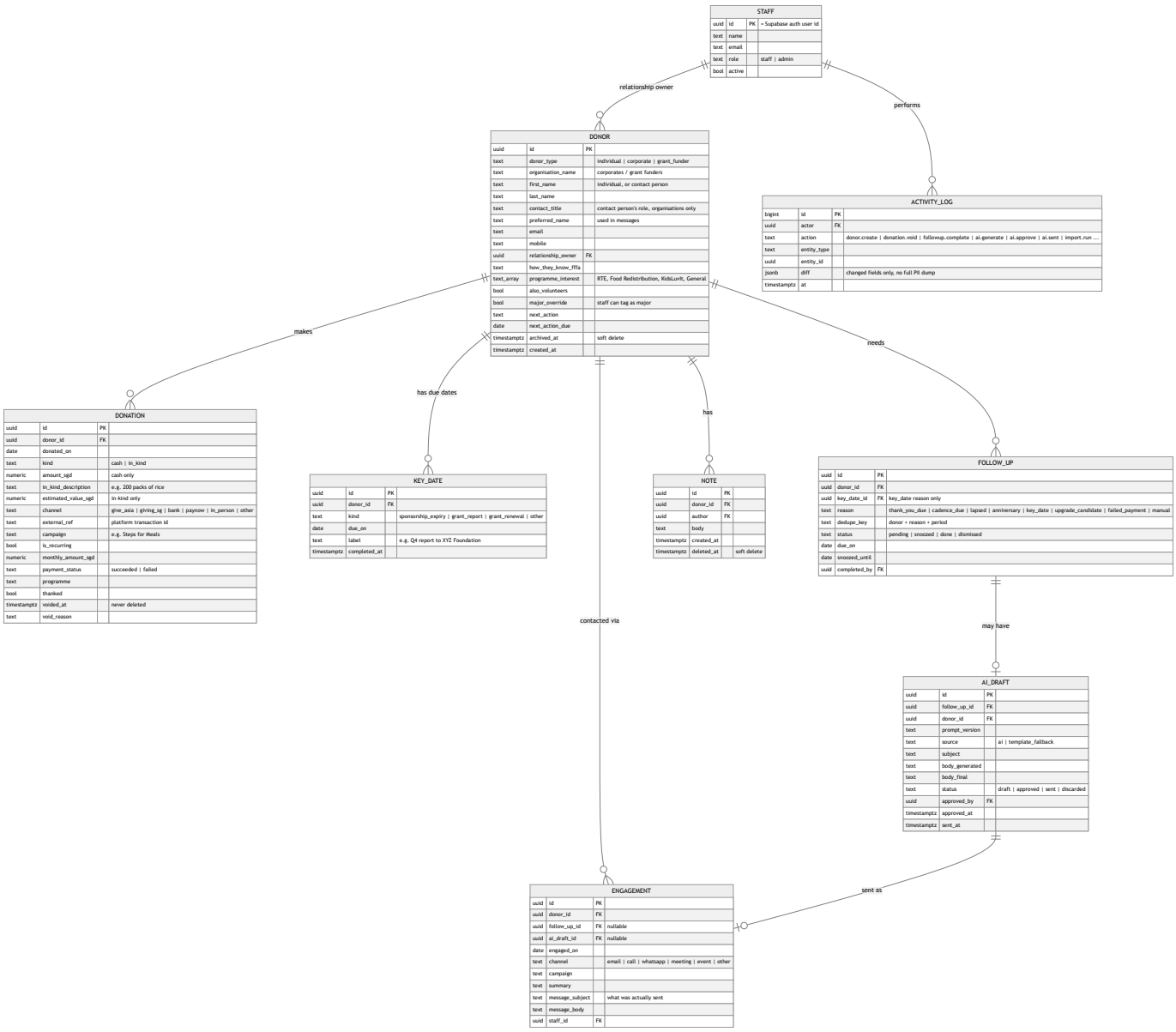
2. Architecture



Key decisions

Decision	Choice	Why	Alternatives considered
Frontend	Keep React 19 + Vite + TS + Tailwind from the current repo	Team already has it; fast to build	Next.js (more setup, no SSR need)
Backend	Keep a thin Express API in front of the DB	Gemini key must stay server-side; rules, logging and AI live in one place; client never touches the DB directly	Client → Supabase directly with Row Level Security (less code, but business rules and audit logging get scattered and the AI key still needs a server)
Database	Supabase Postgres	Relational data (donors ↔ donations ↔ follow-ups), free tier, table editor staff can open in an emergency, backups	Firestore (weaker for joins/aggregates), SQLite (no managed hosting/auth)
Auth	Supabase Auth (email + password or magic link) + server-side staff allowlist	Built-in, no custom password handling	Google Workspace SSO (depends on FFFA's setup — revisit in Phase 2)
AI model	Gemini Flash via `@google/genai` (exact model id verified at build time)	Already wired in repo, cheap, structured JSON output	Claude / GPT (equally viable; switching is a single service file)
Follow-up queue	**Materialised**: rules write `follow_ups` rows; daily job + on-demand refresh	Gives each follow-up a status/history (DM4 "removes from pending"), lets staff snooze/complete, and the dashboard can count them cheaply	Compute on read (simpler, but no status tracking or history)
Hosting	Container on Cloud Run (or Render) — decide with FFFA	FFFA has hosting funding; container keeps us portable	Vercel (fine for SPA, awkward for the daily job)

3. Data model



`donors.key_date` / `key_date_label` from v1.1 move into the `KEY_DATE` table, so a grant funder can have a report deadline **and** a renewal date. `donations.event` becomes `campaign`.

Organisations. A corporate or grant funder is one donor record: `organisation_name` is the donor, and first/last name, `contact_title`, email and mobile describe the contact person. Messages greet the contact person by name. Campaign + organisation together support FFFA's campaign reporting and grant matching.

In-kind donations (food, essentials) are donations with `kind = in_kind`, a description and an estimated value. They show on the profile and in reports, and they get thank-yous, but tiers and the major rule count **cash only**.

Computed, never stored: a `donor_summary` view derives these on read so they can't go stale: last donation (date, amount, campaign), last engagement (date, channel), average donation amount, average days between donations, 12-month cash giving, years with FFFA, **tier**, **is_major** and **days since last engagement**. Failed payments and voided donations are excluded from all totals.

Major donors (FFFA rule). `is_major` = 12-month cash giving ≥ S$5,000 **or** first gift ≥ 3 years ago **or** `major_override` set by staff. Both thresholds live in `org_settings`. Major donors get personal outreach from staff: they still appear in the queue, but AI drafting is disabled for them (PRD: "major and long-standing donors get a human").

Tiers (internal only) stay separate from "major": `tier_settings` holds Bronze / Silver / Gold / Diamond with a 12-month giving threshold and a **contact cadence in days** per tier, used by the `cadence_due` rule. Tier is calculated, never set by hand, and never sent to the AI or

shown to donors.

Other config tables: `rule_settings` (thresholds, editable by admin) and `org_settings` (impact unit **S$10 = 1 meal for one elderly person**, major-donor thresholds, org signature).

Data guardrails (PRD §3): no NRIC/FIN/UEN, no age/income/ethnicity; **dummy data only** (confirmed by FFFA).

Adding NRIC/UEN later (for IRAS tax deduction receipts) needs no rebuild: a separate `tax_identities` table (donor_id, id_type NRIC/FIN/UEN, encrypted value, consent date) added by its own migration, readable by admins only, never returned by donor list/profile endpoints and never sent to the AI. Donation records already carry date, amount, kind and channel, which an IRAS export needs.

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4. Follow-up rules engine (DM9)

Runs daily (cron) and on demand ("Refresh queue" button). Each rule is a pure function `(donor, donations, key dates, settings, today) → FollowUpCandidate | null`, so rules are easy to unit-test and to add later.

Rule	Fires when	Default threshold	Priority
`thank_you_due`	A donation (cash or in-kind) has `thanked = false` and is ≥ N days old	N = 3 days	High
`failed_payment`	A recurring donor's latest monthly payment has `payment_status = failed`	Within 7 days of the failure	High
`key_date`	A `KEY_DATE` (sponsorship expiry, grant report, grant renewal) is due within N days and not completed; one follow-up per key date	N = 30	High
`cadence_due`	Days since last engagement ≥ the donor's tier cadence (going cold)	Per tier in `tier_settings`	High
`lapsed`	Recurring donor with no successful donation for N days, or one-off donor with none for M days	N = 45, **M = 180 (6 months, FFFA)**	High
`anniversary`	First-donation anniversary within the next N days	N = 7	Medium
`upgrade_candidate`	Recurring at the same monthly amount for ≥ 12 months, not asked in the last 12 months	12 months	Low

Thresholds confirmed by FFFA on 20 Oct ("defaults are fine", except one-off lapse at 6 months).

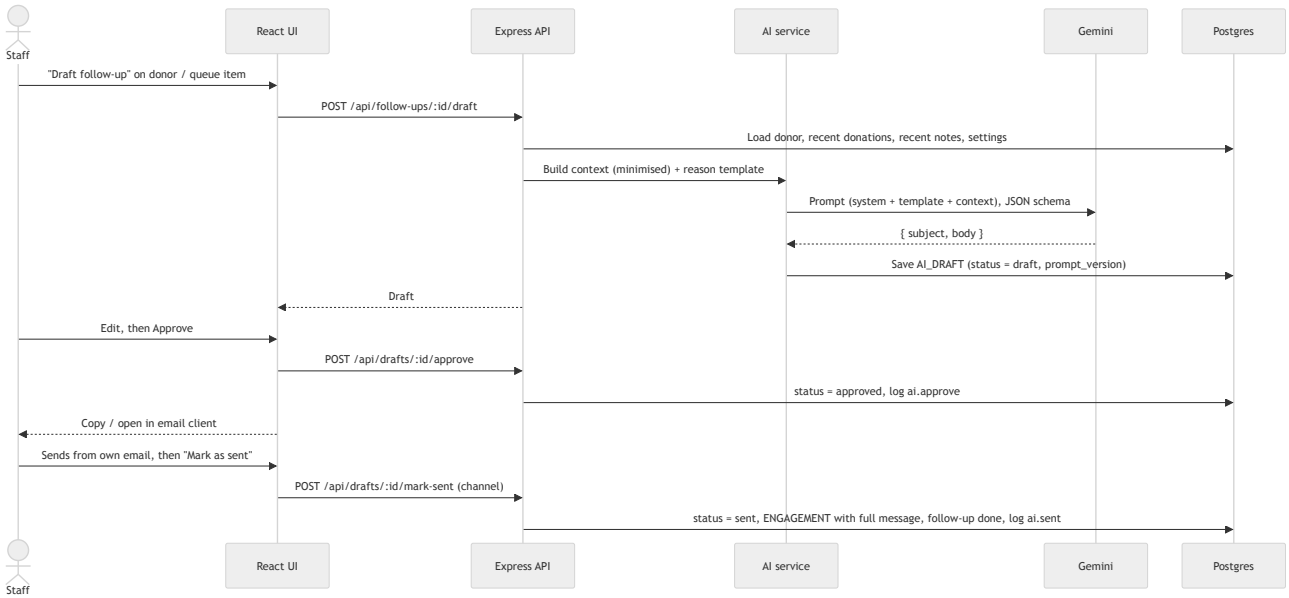
Idempotency: each candidate has a `dedupe_key` (e.g. `donor:123|lapsed|2026-10`, or `keydate:456` for key dates). The engine upserts on that key, so re-running never duplicates and a completed follow-up is not recreated in the same period.

Completing a follow-up records an `ENGAGEMENT` (date, channel, summary, and the message if one was sent), which resets that donor's cadence clock. Completing a `key_date` follow-up can also mark the key date done.

Major donors: still appear in the queue, tagged "personal outreach", with the AI draft button disabled.

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5. AI drafting flow (DM10)



- **The sent message stays with FFFA** (FFFA feedback): "Mark as sent" saves the exact subject and body on an engagement, so the full conversation history lives on the donor's profile, not only in a staff inbox. Messages staff write themselves (e.g. to major donors) can be logged the same way through `POST /api/donors/:id/engagements`.
- **Prompt templates per reason** (thank-you, failed payment, key date, check-in, lapsed, anniversary, upgrade ask), versioned in code (`prompt_version`) so we can compare outputs.
- **Context minimisation:** the model gets the display name (contact person for organisations), donor type, organisation name, tenure, donation amounts/dates/campaign/programme, in-kind descriptions, relationship notes, the impact unit and the reason. **Never** email, mobile, tier or tax identifiers; notes are scrubbed of emails, phone numbers and NRICs.
- **Impact unit:** S$10 = 1 meal for one elderly person. Meal counts and upgrade figures are computed in code (e.g. "S$30 → S$45 a month = 1 more meal each month"), never by the model.
- **Failed payments:** a short, low-friction note with no guilt, asking them to update their card details on the platform they give through.
- **Tone guardrails** in the system prompt: warm, specific, no guilt, at most one modest ask, Singapore English, FFFA voice.
- **Nothing sends from the system.** Sending stays with staff (PRD out of scope); FFFA confirmed they like that "someone on the team always reviews and sends every message".
- **Failure mode:** if Gemini fails or times out, fall back to the static template for that reason (`source = template_fallback`) so staff are never blocked.
- **Evaluation:** a small fixed set of ~10 dummy donors × each reason; we review outputs for tone, accuracy (no invented facts) and length each time prompts change.

6. API (v1)

All routes require a valid staff session. Writes go through the activity logger.

Method	Route	PRD
GET	`/api/donors?query=&type=&tier=&lastDonationBefore=&owner=`	DM1, DM2
POST / PATCH	`/api/donors` , `/api/donors/:id`	DM1
GET	`/api/donors/:id` (profile: summary, donations, engagements, notes, follow-ups, drafts)	DM3
POST / DELETE	`/api/donors/:id/notes` , `/api/notes/:id` (soft delete)	DM3
POST	`/api/donors/:id/donations` (cash or in-kind), `/api/donations/:id/void`	DM8
POST	`/api/donors/:id/engagements`	DM3, DM4
GET	`/api/follow-ups?status=pending` (the "Today" queue)	DM9
POST	`/api/follow-ups/refresh`	DM9
POST	`/api/follow-ups/:id/complete` (creates an engagement) · `/snooze` · `/dismiss`	DM4
POST	`/api/follow-ups/:id/draft` · `/api/drafts/:id/approve` · `/api/drafts/:id/mark-sent` · `/api/drafts/:id/discard`	DM10
GET / POST / PATCH	`/api/donors/:id/key-dates` , `/api/key-dates/:id` (incl. mark complete)	DM9
GET	`/api/dashboard` (aggregates + monthly series)	DM5
POST	`/api/import/preview` · `/api/import/commit` (CSV; see import matching below)	DM6
GET / PATCH	`/api/settings` (admin: rules, tiers, impact unit)	DM9, DM10
GET	`/api/activity?entity=` (admin only)	DM7
POST / DELETE	`/api/donors/:id/archive` · `DELETE /api/donors/:id` (admin only, PDPA requests)	DM1

Delete semantics. Each resource gets the operation its data needs rather than a blanket DELETE:

Resource	Operation	Why
Donor	Archive (soft); hard delete admin-only	PDPA requests to remove personal data
Donation	Void with reason, never deleted	Financial record; feeds a later IRAS export
Note	Soft delete (author or admin)	Recoverable mistakes
Follow-up	Dismiss	Keeps queue history
AI draft	Discard	Keeps generation history for evaluation
Activity log	Never	Audit trail

Import matching (Giving.sg and GIVE.asia, FFFA feedback). The same person often gives on both platforms. On import:

1. **Donations** are de-duplicated by `(channel, external_ref)` , so re-importing the same export never double-counts.
2. **Donors** are matched by normalised email, then by normalised mobile (+65 stripped, digits only).
3. Rows that match on name only, or match two different donors, are **not merged automatically**: the preview lists them for staff to confirm "same person" or "new donor".
4. Matched donations attach to the existing donor with their own `channel` , so the profile shows giving across both platforms.

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7. Screens

1. **Login**
2. **Today** — pending follow-ups grouped by priority; actions: draft, complete, snooze. *(The demo's home screen.)*

3. **Donors** — table with search and filters.
4. **Donor profile** — details, donations, notes timeline, follow-ups, "Log donation", "Draft follow-up".
5. **Dashboard** — total/active donors, donors by tier, pending follow-ups, upgrade candidates, plus two monthly charts: **donations** and **outbound engagements** (from the engagements table, not the activity log). Aggregates only; built after the core flow.
6. **Import** — upload CSV, preview with validation errors, commit.
7. **Settings** (admin) — rule thresholds, tier thresholds and cadences, impact unit, staff allowlist.

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8. Security (DM7, PRD §3)

- Supabase Auth login; the API verifies the JWT on every request **and** checks the user is in the `staff` table and active.
- Roles: `staff` (day-to-day), `admin` (settings, staff list, activity log).
- The browser never gets DB credentials; Postgres Row Level Security set to deny-all for the anon key, server uses the service role.
- Secrets (`GEMINI_API_KEY` , Supabase service key) only in server env.
- Activity log for every write and every AI generate/approve (who, what, when, changed fields).
- Dashboards and exports show aggregates only; no bulk PII export in Phase 1.
- Dummy data in all environments (confirmed by FFFA, 20 Oct).
- NRIC/UEN, when added later, go in a separate encrypted, admin-only table (section 3).

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9. Implementation status

Done (merged)	Next
Schema `001_init.sql` + `002_fffa_feedback.sql` (campaign, contact title, major override, in-kind, payment status, external ref, key dates, sent messages, S$10 impact unit, 6-month lapse)	Donor / donation / key-date API, rules engine (Swaraj)
DB layer (Postgres / PGlite) with transactions, auth hook, activity log, shared types	Supabase auth + screens (Chris)
DM10: draft / approve / mark as sent / discard, major-donor block, failed-payment template, prompt `dm10-v2`	Seed data, import matching, dashboard (Lynn)
Old AI Studio UI kept on `legacy/ai-studio-demo`	AI evaluation set once Gemini key is available

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10. Workstream ownership (Systems)

Estimates are in ideal working days. v1.2 adds about 4 days of work from FFFA's feedback, spread so the load stays even (activity log moves from Swaraj to Lynn).

Person	Owns	PRD	Days
Ghanashyam (lead)	Architecture, schema/types, migration 002 (1.5) · AI drafting incl. mark-as-sent, major block, impact unit, failed-payment template (3) · AI evaluation with Lynn (0.5) · integration, stand-ups, demo prep (1.5)	DM10	~6.5
Swaraj	Donor, donation (cash + in-kind) and notes API (2.5) · rules engine incl. key dates and failed payments, follow-up API (3.5) · key dates API (0.5)	DM1–DM4, DM8, DM9	~6.5
Chris	Auth and staff access (1.5) · screens: Login, Today, Donors, Donor profile (incl. organisation contact, in-kind, key dates, conversation history), Import, Settings (5.5)	DM7 (auth), DM1–DM4 (UI)	~7
Lynn	Seed data (0.5) · CSV import incl. cross-platform matching (2.5) · dashboard API and screen (1.5) · activity log read endpoint (1) · AI evaluation set (0.5) · test checklist (1)	DM5, DM6, DM7 (log), DM10 (eval)	~7

Ways of working

- **Day 1, together:** agree the shared schema, TypeScript types and API conventions before building, so everyone can work in parallel.

- **Peer code review:** every pull request is reviewed by one other teammate; reviewers rotate so everyone understands every part of the system.
- **Short stand-ups** (async in the group is fine): done, next, blocked.
- **Team demos:** the lead opens and closes; each owner presents the part they built.

People & Processes (Fiona's team) provides: FFFA field confirmation, rule thresholds, impact unit, tone guidance and tester names.

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11. Milestones

Date	Milestone
19 Oct	✅ Platform foundation + DM10 draft/approve merged (PR #1)
20 Oct	Design v1.2 with FFFA feedback; PRD v3 draft to Fiona
21 Oct	**Session 2: recalibrate as a team and build** (walk through v1.2, confirm owners, start slices)
23 Oct	Migration 002; donors/donations/notes API; rules engine core; Today + profile screens on seed data
25 Oct	Full flow: queue → draft → approve → mark as sent → history on profile; import + dashboard
26–27 Oct	FFFA testing (tester confirmed), fixes, staff guide, handover notes
28 Oct	**Final demo and handover**

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12. Confirmed decisions, assumptions and open questions

Confirmed by FFFA (20 Oct)

- Donor fields as listed, plus campaign and organisation contact person.
- Major donor: S$5,000+ a year or 3+ years with FFFA, or tagged by staff.
- Rule defaults fine; one-off donors lapse after 6 months.
- Tester: FFFA's respondent. Data: dummy only. Impact unit: S$10 = one meal for one elderly person.

Assumptions (we build on these; FFFA can correct them anytime)

- AI drafts: any staff user can approve; major donors are excluded from AI drafting.
- Email: staff send from their own email client and then mark the message as sent.
- In-kind donations count towards thank-yous and reports, not towards tiers or major status.
- Hosting and post-handover ownership: decided with FFFA before 28 Oct; the build stays container-portable.

Question sheet (answered by FFFA's email, 20 Oct)

1. *Other donor data such as demographics (age, income, ethnicity, dialect) or preferences?* Answered: the listed fields work, plus campaign and organisation contact person. No demographic fields; this also matches PRD §3.
2. *Operations dashboard (active donors, beneficiaries, inventory)?* Not needed for Phase 1: the dashboard covers donor figures only (DM5), with donations by campaign added for FFFA's campaign reporting.

Still open: hosting and who owns and maintains the system after handover (before 28 Oct).